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# Beyond V2X: A Review of Large Language Models in In-Vehicle Autonomous Driving Systems

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## Abstract

Large Language Models (LLMs) are emerging as key enablers of intelligent in-vehicle systems, supporting conversational interfaces, diagnostics, and decision support in autonomous driving. While prior surveys have focused on LLMs in vehicular infrastructure, little attention has been paid to their integration within the in-vehicle environment-where safetycritical decisions occur. This paper offers a comprehensive review of LLM applications inside vehicles, presenting a structured taxonomy and analyzing recent developments across academia and industry. We identify key deployment challenges, including latency, determinism, real-time reliability, explainability, and compatibility with legacy automotive protocols. In parallel, we highlight critical security and privacy threats such as prompt injection, adversarial manipulation, data leakage, and model inversion, which are amplified in over-the-air and safety-critical contexts. We further argue that unsafe in-vehicle LLM behavior can propagate into the broader Intelligent Transportation System (ITS), increasing systemic risk. To mitigate these concerns, we propose a targeted research agenda and evaluation framework for LLMs in vehicular settings. Our goal is to bridge a key gap in existing literature and guide the secure, robust, and humanaligned deployment of LLMs in automotive systems.

## Keywords

adversarial attacks, autonomous driving, communication, networking and broadcast technologies, data leakage, edge ai, human-machine interface (hmi), in-vehicle networks, intelligent transportation systems (its), large language models (llms), privacy, prompt injection, safety-critical systems, secure ai deployment, transportation, vehicle diagnostics

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# Beyond V2X: A Review of Large Language Models in In-Vehicle Autonomous Driving Systems

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***Abstract*—Large Language Models (LLMs) are emerging as key enablers of intelligent in-vehicle systems, supporting conversational interfaces, diagnostics, and decision support in autonomous driving. While prior surveys have focused on LLMs in vehicular infrastructure, little attention has been paid to their integration within the in-vehicle environment—where safety-critical decisions occur. This paper offers a comprehensive review of LLM applications inside vehicles, presenting a structured taxonomy and analyzing recent developments across academia and industry. We identify key deployment challenges, including latency, determinism, real-time reliability, explainability, and compatibility with legacy automotive protocols. In parallel, we highlight critical security and privacy threats such as prompt injection, adversarial manipulation, data leakage, and model inversion, which are amplified in over-the-air and safety-critical contexts. We further argue that unsafe in-vehicle LLM behavior can propagate into the broader Intelligent Transportation System (ITS), increasing systemic risk. To mitigate these concerns, we propose a targeted research agenda and evaluation framework for LLMs in vehicular settings. Our goal is to bridge a key gap in existing literature and guide the secure, robust, and human-aligned deployment of LLMs in automotive systems.**

***Index Terms*—Large Language Models (LLMs), In-Vehicle Networks, Intelligent Transportation Systems (ITS), Autonomous Driving, Human-Machine Interface (HMI), Edge AI, Safety-Critical Systems, Vehicle Diagnostics, Prompt Injection, Privacy, Secure AI Deployment, Adversarial Attacks, Data Leakage**

## I. Introduction

Large Language Models (LLMs) have emerged as transformative tools across domains, enabling machines to understand, generate, and interact using natural language with near-human fluency. Their adoption spans sectors ranging from healthcare to software engineering, and their increasing accessibility via APIs and open-source models has rapidly accelerated their deployment in real-world systems [1]–[3]. Within the automotive industry and the broader landscape of Intelligent Transportation Systems (ITS), LLMs are beginning to play a crucial role—particularly in user interaction, diagnostics, navigation, and support for autonomous driving.

Modern vehicles are no longer simple mechanical platforms; they are complex cyber-physical systems capable of processing hundreds of sensor inputs and running sophisticated AI models. A typical autonomous vehicle generates over 25GB of data per hour and comprises more than 100 million lines of code [4], [5]. As systems become more software-defined, automakers and mobility providers are exploring ways to integrate LLMs as onboard agents to enhance perception, planning, and human-machine interaction. For example, Mercedes-Benz recently partnered with OpenAI to integrate ChatGPT into its MBUX infotainment system [6], aiming to create a more natural conversational interface. Tesla's software updates suggest the growing use of NLP pipelines for voice control and command parsing. Companies like Zoox and Cruise are reportedly exploring AI copilots to assist autonomous taxis with real-time user interaction [7].

Despite these advances, the academic and industrial research community has so far placed disproportionate focus on LLM applications in connected vehicular networks—such as vehicle-to-everything (V2X) communication, traffic summarization, and infrastructure-assisted driving [8]. While important, this leaves a significant gap in understanding the role, design, and risks of deploying LLMs inside the vehicle itself, particularly in the decision loops of autonomous driving systems. Unlike V2X tasks where LLMs often serve as offline processing tools or cloud-based summarizers, in-vehicle LLMs face fundamentally different constraints: limited compute, real-time latency, safety-critical decision impact, and deep integration with legacy automotive stacks like CAN, FlexRay, and ISO 26262-compliant components.

Moreover, the deployment architecture and role of LLMs inside the vehicle remains poorly defined. Should the LLM act as a co-pilot, only assisting high-level interpretation and interaction? Or should it directly participate in real-time decision-making, perception summarization, or path planning? These questions have profound implications—not only for system design and hardware requirements—but also for safety, accountability, and regulation. Unlike deterministic perception or control modules, LLMs can hallucinate, fail unpredictably, or respond to adversarial prompts, making them difficult to certify under current automotive safety standards.

The use of LLMs in the in-vehicle environment also introduces novel security and privacy threats. Prompt injection, adversarial manipulation, and reverse engineering attacks—well-documented in open-domain LLMs—could now directly influence steering, braking, or verbal driver feedback if not properly isolated. Furthermore, privacy risks emerge when user voice commands, location history, or vehicle diagnostics are processed without robust safeguards, especially in vehicles with over-the-air (OTA) update capabilities. These vulnerabilities can create attack surfaces not only for individual vehicles but also for entire fleets connected to central LLM-assisted orchestration systems [9], [10].

In this paper, we aim to bridge a critical gap in the literature by providing a focused and in-depth review of the use of LLMs in in-vehicle systems, distinct from prior work centered on V2X or cloud-based ITS applications. Our goal is not only to consolidate current understanding but also to clarify the architectural ambiguities that arise when deploying LLMs within safety-critical, real-time vehicular environments. By analyzing recent research and emerging deployments, we construct a taxonomy of in-vehicle LLM functionalities, assess key limitations, and examine the cascading impact of LLM behavior on broader transportation infrastructure. Furthermore, we emphasize the unique security and privacy implications introduced by LLM integration at the vehicle level and propose a set of evaluation metrics and future research priorities to guide the development of trustworthy and certifiable LLM-based vehicular systems.

To summarize, the key contributions of this paper are as follows:

- We present a focused and comprehensive review of LLM applications within **in-vehicle systems**, distinguishing our work from prior surveys that focus primarily on connected ITS or infrastructure-level use cases.
- We propose a structured taxonomy of **in-vehicle LLM applications**, categorizing use cases by functional role (e.g., copilots, diagnostics, decision support) and integration depth.
- We analyze the **architectural ambiguity** in deploying LLMs inside the vehicle, highlighting design uncertainties around agent roles, inference locality, and system-level interaction.
- We identify key **technical, safety, and regulatory challenges**, including latency, hallucination risks, prompt injection, and the difficulty of applying existing automotive certification standards (e.g., ISO 26262) to LLM-based components.
- We discuss how **LLM vulnerabilities may propagate system-wide**, potentially affecting V2X communication and ITS infrastructure through misaligned onboard behavior.
- We propose a set of **future research directions**, emphasizing safety, interpretability, and secure integration of LLMs in autonomous vehicles.

## II. Organization of the Paper

The remainder of this paper is organized as follows. Section III presents the necessary background on Large Language Models (LLMs), the architecture of Intelligent Transportation Systems (ITS), and in-vehicle communication protocols such as CAN. Section IV examines emerging opportunities for integrating LLMs within vehicles, focusing on applications such as contextual reasoning, diagnostics, and voice-based interaction. Section V introduces a taxonomy of LLM applications in in-vehicle systems, classifying them by integration level, modality, and functional role.

Section VI analyzes the security and privacy implications of in-vehicle LLM deployments, including risks such as prompt injection, model inversion, and data leakage. Section VII discusses additional challenges beyond security and privacy in applying LLMs to the in-vehicle domain. Section VIII reviews existing efforts to integrate LLMs within vehicles, with particular emphasis on works addressing security and privacy concerns, focusing primarily on studies situated in the in-vehicle context. Section IX identifies gaps in the current literature regarding LLM deployment in vehicles. Section X outlines promising future research directions for LLM integration in the in-vehicle domain. Finally, Section XI concludes the paper with a summary of key findings and recommendations for future work.

## III. Background

### A. Large Language Models: Capabilities and Evolution

Large Language Models (LLMs) are deep neural networks trained to predict and generate sequences of text by learning statistical relationships over massive corpora. Built primarily upon transformer architectures [11], LLMs such as GPT-3, GPT-4 [1], LLaMA [2], and PaLM [12] have demonstrated exceptional capabilities in language understanding, summarization, reasoning, and multi-turn dialogue. These models typically consist of billions of parameters and are pre-trained on internet-scale data using next-token prediction objectives. Subsequently, many undergo fine-tuning via reinforcement learning with human feedback (RLHF) [13] or instruction tuning to follow natural language prompts more effectively.

LLMs are inherently multimodal-capable and can be extended to understand structured data, sensor descriptions, images, and code. Their strengths include context-sensitive interpretation, zero-shot generalization, and dynamic response generation. These traits make them appealing for deployment in interactive systems, including voice assistants, code analysis tools, and increasingly, intelligent vehicles. However, they also carry risks such as non-determinism, hallucinations (producing plausible but incorrect outputs), and susceptibility to adversarial prompts and model inversion attacks [9], [10].

### B. Intelligent Transportation Systems Overview

Intelligent Transportation Systems (ITS) represent the convergence of embedded electronics, communication protocols, and control systems to improve the safety, efficiency, and adaptability of modern transportation infrastructure. At the core of this ecosystem lies the in-vehicle system, responsible for sensing the environment, processing control logic, and coordinating actuation across key components such as steering, braking, and acceleration.

Modern vehicles comprise a network of Electronic Control Units (ECUs), each dedicated to specific subsystems including the powertrain, climate control, infotainment, advanced driver-assistance systems (ADAS), and more. These ECUs communicate using in-vehicle networks (IVNs) designed to support real-time, fault-tolerant, and priority-based message delivery.

The most widely adopted protocol in IVNs is the Controller Area Network (CAN), developed by Bosch in the 1980s and standardized as ISO 11898 [14]. CAN is a broadcast, message-based protocol in which each frame is identified by a unique arbitration ID. Rather than source-destination

addressing, all messages are broadcast on the shared bus, and each ECU determines whether to process a message based on its identifier. The standard CAN message frame includes an 11-bit (or extended 29-bit) identifier, a Data Length Code (DLC), and up to 8 bytes of payload. Arbitration is achieved through a non-destructive bitwise comparison, favoring lower ID values as higher priority. A digrammatic representation of the message arbitration mechanism has been shown in Figure 1.

CAN's strengths lie in its simplicity, low-latency communication, robustness under noisy environments, and graceful fault handling. These qualities have made it the standard backbone for safety-critical subsystems such as engine control, braking, and transmission.

However, modern vehicle architectures are evolving. The increasing data throughput required by sensors like LiDAR, radar, and high-definition cameras—alongside infotainment demands—has prompted the adoption of supplementary protocols. These include:

- **LIN (Local Interconnect Network):** A low-cost, single-master protocol used for non-critical subsystems like mirrors, windows, or interior lighting.
- **FlexRay:** A time-triggered protocol offering higher bandwidth and deterministic communication, often used in safety-redundant systems such as x-by-wire applications.
- **Automotive Ethernet:** A high-bandwidth, switched protocol suitable for transmitting large volumes of data, such as camera feeds or OTA updates.

In parallel, vehicle architectures are transitioning toward centralized domain controllers and zonal networks to simplify wiring, improve scalability, and support software-defined functionality. Standards like AUTOSAR (Automotive Open System Architecture) ensure interoperability among components and support the design of functionally safe systems in accordance with ISO 26262 guidelines.

While these networks and standards have proven effective for traditional rule-based control logic, the increasing software complexity of modern vehicles places new demands on communication architectures. Maintaining determinism, low latency, and fault isolation remains central to the safe operation of next-generation transportation systems.

## IV. Application Use Cases of LLMs in In-Vehicle Networks

While most prior works exploring LLMs in transportation focus on cloud-based or connected vehicle environments, their integration directly within in-vehicle networks presents transformative possibilities. LLMs can act as reasoning layers between human input, sensor data, and vehicle actuation—enabling richer interactions, situational explanations, and context-aware decision support. Below, we outline several representative use cases that exemplify this emerging capability.

### *A. Voice-Based Contextual Assistance and Diagnostic Feedback*

Modern vehicles are increasingly equipped with voice assistant interfaces [15]; however, these systems are largely rule-based and limited to predefined command sets. LLMs can extend these capabilities by understanding open-ended, ambiguous, or compound queries.

> *"Why did you brake suddenly just now?"*
> *"Can I still drive with the current engine warning light?"*

An in-vehicle LLM can access CAN data streams to infer context from the brake system ECU, accelerometer data, and road conditions, and respond:

> *"The emergency braking was triggered due to sudden deceleration of the vehicle ahead, detected by the radar sensor at 3.4 meters."*

This enables explainable automation and enhances trust in semi-autonomous systems. Voice queries can also be extended to cover topics such as congestion, estimated time of arrival (ETA), and safe rerouting based on real-time telemetry and map APIs. Integration with lightweight speech recognition systems, such as Whisper or embedded ASR engines, is feasible on modern automotive SoCs [1], [16].

Interpreting vehicle diagnostics (e.g., DTCs, warning lights) remains non-trivial for non-experts. LLMs fine-tuned on vehicle manuals, service logs, and OBD-II codes can provide natural language explanations. For instance, if a P0420 (catalyst efficiency below threshold) code is raised, the LLM may respond:

> *"The vehicle has detected reduced efficiency in the catalytic converter, which may be caused by a faulty oxygen sensor or exhaust leak. You can continue driving, but emissions may be affected."*

This use case can be further extended to generate **technician-readable summaries** of historical error patterns and sensor failures, potentially streamlining service workflows. Additionally, it brings transparency to the repair process by enabling the technician's diagnosis and repair work to be relayed back to the customer in layman's terms, fostering greater trust. The LLM can compare pre- and post-repair telemetry to identify which subsystems were underperforming and how their behavior changed after servicing, enhancing both user satisfaction and service accountability.

### *B. Real-Time Decision Support for Drivers*

By integrating sensor fusion data—including LiDAR, radar, GPS, and ADAS feeds—Large Language Models (LLMs) can serve as intelligent copilots offering real-time advisory input. For example:

> *"There's heavy fog ahead. Do you want to reduce speed or enable lane-keeping assist?"*
> *"Your battery range is tight for the planned trip. There's a charging station in 8 miles—should I reroute?"*

These decisions rely on the LLM's ability to interpret real-time sensor values, map context, and historical driver preferences. While control actions remain governed by deterministic modules or certified rule-based systems, the LLM introduces semantic reasoning and just-in-time decision framing to support the driver with minimal human distraction/intervention while driving.

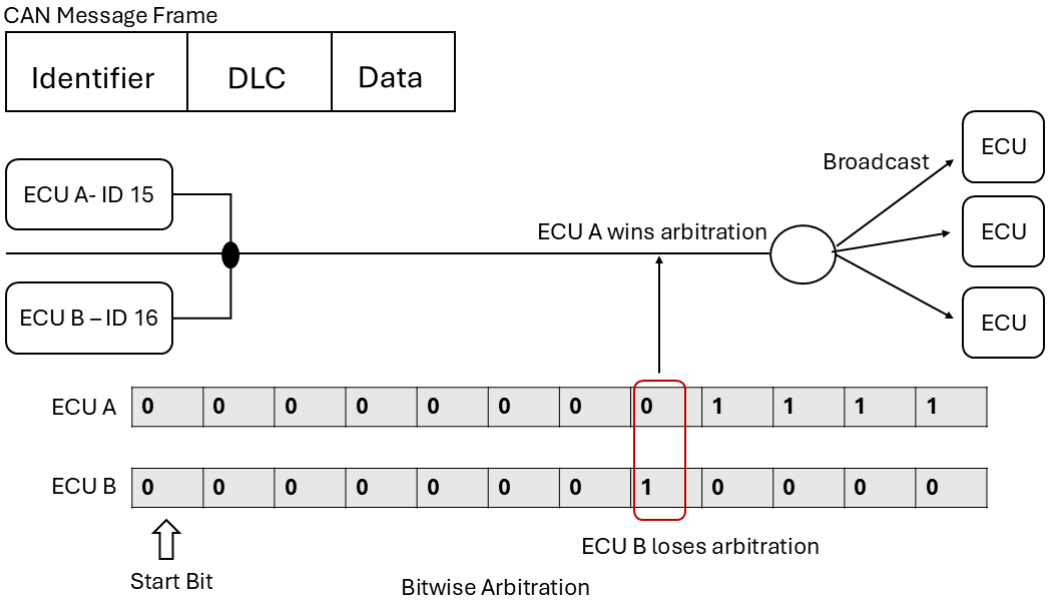



Fig. 1. Message Arbitration Mechanism in CAN: ECU A with identifier 0x0F (binary: 00001111) and ECU B with identifier 0x10 (binary: 00010000) begin transmission simultaneously. Due to the dominant/recessive bit arbitration scheme, ECU B loses arbitration at bit position 4 where it sends a recessive bit (1) while ECU A sends a dominant bit (0). As a result, ECU B stops transmitting, and ECU A wins arbitration and proceeds to broadcast its message to all nodes on the CAN bus.

Beyond conventional dashboard alerts, LLMs can dynamically fuse context from external third-party services such as Waze or Google Maps to deliver higher-level guidance. For example, the LLM can interpret fog or congestion alerts not just as travel warnings, but as conditions requiring immediate action—especially for emergency vehicles or critical logistics. Likewise, trip planning queries can integrate not only state-of-charge data from the battery management system, but also geospatial proximity to gas stations, charging points, dining areas, or rest stops—thereby creating a proactive, multi-intent co-driving experience tailored to the user's situation and preferences.

### C. Personalized Driver Interaction and Feedback

From a design perspective, Large Language Models (LLMs) can be integrated into driver monitoring and personalization systems to enhance both safety and user experience. By interpreting interaction patterns, telemetry data, and behavioral cues, an LLM may detect signs of stress, fatigue, or agitation and offer context-sensitive suggestions:

*"You seem tired—would you like to take a break? There's a rest stop in 5 miles."*

LLMs can adapt explanations to a driver's mood, personalize voice tone, and dynamically modify content delivery, contributing to safer and more engaging in-vehicle experiences. Furthermore, LLMs can analyze behavioral habits—such as average acceleration, hard braking frequency, or lane-switching tendencies—and generate personalized insights:

*"You tend to brake hard during downhill turns—would you like tips for smoother handling?"*

This blends personalization, coaching, and safety augmentation. In the context of new or novice drivers, LLMs can serve as virtual driving instructors, offering guidance in real-time. Beyond coaching, this data can also be used to explain how driving behavior might influence journey safety, comfort, or even insurance outcomes.

Insurance providers often rely on external telematics devices to assess driver behavior for policy adjustment. An in-vehicle LLM can interpret the same telemetry data to:

- Identify and summarize unsafe driving patterns,
- Infer driver mood based on behavior and vehicle dynamics,
- Proactively inform the driver how their behavior could affect the journey or insurance premiums.

This approach introduces transparency into driver evaluation, while also fostering self-improvement through timely and constructive feedback.

### D. In-Vehicle Knowledge Interfaces

Large Language Models (LLMs) can function as **natural language interfaces to vehicle documentation**, allowing drivers to retrieve information without needing to browse static manuals. For example:

*"Where is the fuse box for the AC system?"*
*"How often should I change the transmission fluid?"*

By indexing and embedding vehicle-specific manuals and technical documentation, an in-vehicle LLM can provide concise, accurate responses tailored to the vehicle model and context. This capability reduces friction in accessing important information, especially in urgent or unfamiliar situations.

The approach can be extended to create onboard help systems for explaining over-the-air updates, demonstrating infotainment features, or assisting with configuration settings.

Importantly, LLMs trained or fine-tuned on OEM-provided manuals can offer structured and reliable guidance.

This is particularly beneficial for younger or amateur drivers, who may not be familiar with terminology or maintenance schedules. Instead of consulting dense documentation, they can rely on conversational interactions with the LLM to better understand operational requirements, interpret warnings, and maintain vehicle health, thereby enhancing safety, engagement, and self-sufficiency.

Thus, LLMs can also enable a diverse set of applications depending on their access privileges, reasoning scope, and user interaction modality. Table III categorizes these applications along key dimensions.

Overall, the integration of LLMs into in-vehicle systems presents a unique opportunity to elevate vehicles from reactive automation platforms to conversational, context-aware intelligent agents. However, these opportunities must be weighed against strict latency, safety, and resource constraints, which are explored further in the challenges and architecture sections of this review. A possible architecture for such a LLM enabled voice assistant system architecture has been proposed in Figure 2

## V. Taxonomy of In-Vehicle LLM Applications

To contextualize the role of LLMs in in-vehicle systems, we propose a structured taxonomy based on their integration depth, mode of interaction, and functional scope. This helps differentiate between voice-driven copilots, diagnostic agents, or policy advisors in terms of their deployment model and responsibility level.

LLMs may operate in one or more of the following architectural roles within in-vehicle systems as summarized in Table I:

- **Interface Layer:** Acts as the conversational bridge between driver/passengers and vehicle systems. For example, when the driver asks, "Why is my tire pressure warning on?", the LLM interprets the query, retrieves data from the Tire Pressure Monitoring System (TPMS), and replies, "Your front left tire is at 28 PSI, which is below the recommended 33 PSI."
- **Reasoning Layer:** Synthesizes inputs from sensors, user interactions, and historical data to produce context-aware advice. For instance, it can combine rain sensor data, wheel slip detection, and upcoming sharp turn information from maps to advise: "Reduce speed to 35 mph due to slippery curve ahead."
- **Explanation Layer:** Translates actuator or autonomous driving decisions into human-understandable explanations. For example, after sudden braking, the LLM may explain: "Braking was initiated because the front camera detected a cyclist entering the lane at 18 meters, moving at 15 km/h."
- **Policy Advisor:** Suggests optimized but non-binding decisions for efficiency, safety, or comfort. For example: "Switch to Eco Mode to reach your destination without charging" or "Take Highway 58 to save 12 minutes and extend EV range by 8 miles."

Based on deployment, models can be categorized as:

- **Edge-only:** Lightweight LLMs deployed entirely within the vehicle, suitable for rural or military environments with no connectivity. Example: a quantized LLaMA 3 8B model answering safety queries from local maps and sensors without cloud access.
- **Hybrid:** Inference split between onboard and cloud components. Example: the local LLM compiles trip statistics, which are then summarized and enhanced by a cloud-hosted LLM for personalized driving efficiency recommendations.
- **Cloud-dependent:** Most reasoning is offloaded to cloud APIs. Example: a passenger requests a scenic EV route; the system sends preferences and battery status to a cloud LLM, which returns a detailed itinerary with charging stops and lunch options.

## VI. Security and Privacy Challenges in In-Vehicle LLM Integration

The deployment of Large Language Models (LLMs) within in-vehicle networks introduces new dimensions of vulnerability beyond traditional automotive threats. While conventional ECU-level security focuses on bus access, message integrity, and firmware updates, LLMs operate at the language, logic, and behavior layers—creating novel attack surfaces not previously seen in embedded systems.

### A. Security and Privacy in the Context of In-Vehicle LLMs

In the in-vehicle environment, LLMs may process sensitive inputs such as diagnostic data, driver speech, location history, or real-time sensor values. They might also generate control suggestions, route recommendations, or explanations, which could be acted upon by drivers or external agents. This semantic mediation role makes them a valuable but highly sensitive component in the control loop.

Unlike traditional ECUs that operate on fixed logic and hardcoded thresholds, LLMs are susceptible to prompt manipulation, misalignment, and adversarial inference. Furthermore, their reliance on pretrained knowledge or cloud APIs raises concerns regarding inference privacy and attack entry points through externally accessible channels (e.g., OTA updates, V2X messages).

### B. Threats, Concerns, and Vulnerabilities

We identify several critical security and privacy concerns associated with in-vehicle LLM deployments as summarized in Table II:

- **Prompt Injection and Manipulation:** Malicious users may craft voice or text prompts to manipulate the LLM, hijacking its semantic context. The use of voice assistants, as depicted in Figure 2, is a popular mechanism that can be integrated with in-vehicle LLM models. However, voice interfaces are inherently susceptible to a variety of adversarial attacks. One such example is the *Dolphin Attack* [17], where ultrasonic, inaudible signals are embedded into benign audio to trigger unintended

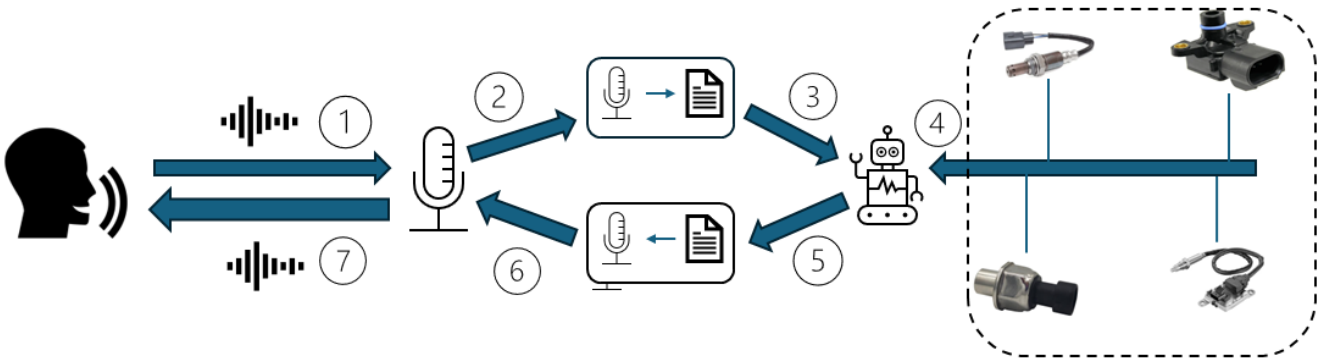



Fig. 2. Proposed architecture for a voice-driven LLM-based in-vehicle assistant. (1) The driver issues a spoken query or command; (2) the speech is captured via an onboard microphone; (3) Automatic Speech Recognition (ASR) converts the voice to text; (4) the LLM module interprets the textual query, optionally accessing real-time sensor data from the in-vehicle network (e.g., via CAN bus or ECUs); (5) the LLM generates a natural language response based on fused sensor-context understanding; (6) the response is passed to a Text-to-Speech (TTS) module; (7) the answer is delivered to the driver in spoken form. This architecture enables hands-free, context-aware interaction between the driver and vehicle systems using natural language and LLM reasoning.

commands. When combined with prompt injection techniques, such as those detailed in [18], this creates a potent vector for manipulating LLM behavior without the driver's awareness.

These threats are especially concerning if the LLM is granted access to in-vehicle control layers—e.g., triggering internal state changes, initiating emergency procedures, or injecting commands into the CAN bus. The severity of such attacks escalates when attackers use multi-turn interaction strategies, like the *SmurfingAttack* [19], to maintain stealth while building malicious context over time. Such strategies can result in remote injection of crafted commands, leading to unauthorized control over vehicular functions and jeopardizing both safety and privacy.

- **Model Inversion and Data Leakage:** Attackers may extract prior interactions, driver behavior, or sensitive routes from model outputs. Such prior information can be exploited to breach the privacy of a driver, particularly when the LLM has access to sensitive contextual data such as GPS coordinates, driving routines, or historical user queries. For instance, repeated access patterns can be used to infer the driver's home, workplace, or frequently visited locations.

  In this context, the work in [20] introduces a *prompt inversion attack*, wherein adversaries aim to recover original user inputs by analyzing the model's internal activations. This demonstrates that even without direct access to stored prompts, malicious participants can infer sensitive content, thereby validating the severity of privacy risks in LLM-powered systems. The implications are particularly critical in vehicular environments, where inferred behavior or routes may reveal not just personal preferences, but patterns exploitable for surveillance, profiling, or physical tracking.
- **Explainability Hallucinations:** Incorrect or fabricated explanations from the LLM may mislead drivers in safety-critical scenarios. LLMs are inherently susceptible to hallucinations—responses that are syntactically plausible but semantically false. LLMs can generate different outputs for the same prompt depending on context history. This vulnerability can severely undermine user trust in automated systems, especially in the context of in-vehicle applications where drivers rely on timely and accurate information.

  The severity of this issue increases under active driving conditions, where the driver may issue voice commands expecting real-time support from the LLM. For instance, if a driver asks, *"Why did the traction control engage?"* and the LLM hallucinates a cause unrelated to the actual sensor data (e.g., "Slippery road detected" when the real reason was a sensor fault), it can lead to confusion or delay in proper action. Similarly, if a driver inquires, *"Can I continue driving with the current warning light?"* and the LLM fabricates a reassuring but incorrect explanation, it could result in system misuse or safety compromise.

  A taxonomy of such hallucinations, particularly in the context of cybersecurity and high-stakes system interaction, has been proposed in [21], where LLM outputs are mapped to confusion matrix components (e.g., false positives, false negatives). Applying such a framework to vehicular LLM deployments is crucial for identifying and mitigating risks where hallucinations may lead to misdiagnosed faults, incorrect warnings, or invalid suggestions.
- **Authentication Bypass via Misalignment:** Ambiguous phrasing or vague intent interpretation may enable access without valid credentials. Researchers have explored the feasibility of authenticating users via voice assistants [22]–[24]. We envision that in the near future, voice-based authentication mechanisms will play a central role in simplifying driver interactions, especially during active driving scenarios.

  Instead of relying on physical interfaces such as touchpads or mobile apps, drivers may authenticate via spoken commands to authorize over-the-air (OTA) updates, execute in-vehicle functions (e.g., opening the moonroof,

activating wipers), alter navigation routes by accessing GPS data, or even start the vehicle itself. This shift toward seamless, hands-free interaction carries both usability advantages and emerging risks.

In adversarial contexts, attackers may exploit LLMs and voice interfaces using inaudible voice injection techniques—such as the Dolphin Attack—and advanced *voice stitching* methods [25], where authentic voice segments are repurposed to fabricate valid-sounding commands. When paired with prompt injection, these methods can bypass voice-based authentication and trigger high-privilege actions without driver consent.

TABLE I
SUMMARY OF LLM ROLES AND DEPLOYMENT ARCHITECTURES IN IN-VEHICLE SYSTEMS

| Category | Example |
|---|---|
| Interface Layer | Answers "Why is my tire warning on?" by retrieving TPMS data and explaining low pressure. |
| Reasoning Layer | Combines rain sensor, wheel slip, and map data to advise slowing down for a slippery curve. |
| Explanation Layer | Explains sudden braking: cyclist detected at 18 m, moving at 15 km/h. |
| Policy Advisor | Suggests Eco Mode or alternate route for time and range savings. |
| Edge-only | Quantized LLaMA 3 8B runs fully onboard, no internet required. |
| Hybrid | Local LLM processes trip stats; cloud LLM generates enhanced recommendations. |
| Cloud-dependent | Cloud LLM plans scenic EV route with chargers and lunch stops. |

## VII. OTHER CHALLENGES IN IN-VEHICLE LLM DEPLOYMENT

Apart from the security challenges, deploying the LLM models inside vehicles introduces a set of unique challenges that span technical, operational, and regulatory dimensions. Unlike conventional cloud-based use cases, in-vehicle deployments require strict guarantees around safety, timing, resource efficiency, and explainability. Below, we outline the key obstacles impeding the widespread adoption of LLMs in automotive systems.

**1) Real-Time and Latency Constraints:** Many in-vehicle tasks—such as diagnostics, driver assistance, or human interaction—require low-latency processing with response times measured in milliseconds. However, LLM inference is computationally intensive and may require significant memory and processing resources. Running such models on embedded hardware is non-trivial, especially given power and thermal constraints. While edge-optimized LLMs and quantization techniques offer some relief, maintaining consistent real-time behavior remains a major challenge.

**3) Lack of Formal Safety Guarantees:** Traditional automotive components are certified under standards like ISO 26262, which require rigorous verification and validation. LLMs, however, operate as black-box models with no formal assurance methods. This makes it difficult to integrate them into safety-critical pathways unless they are sandboxed, monitored by rule-based fallback mechanisms, or wrapped with formal safety envelopes.

**4) Limited Transparency and Debuggability:** In the event of incorrect output or failure, tracing the root cause in an LLM is difficult due to its opaque internal reasoning. Unlike classical rule-based systems, LLMs lack interpretable decision paths, complicating debugging and forensic analysis. This limits their trustworthiness and poses challenges for liability assessment in case of incidents.

**5) Hardware and Integration Overhead:** High-parameter models often require specialized hardware (e.g., GPUs, NPUs), which may not be present in all vehicle configurations. Even with distillation and pruning, running inference onboard demands architectural redesign, including memory management, bandwidth allocation across sensors and actuators, and synchronization with real-time operating systems. These challenges are further amplified in older or vintage vehicles, which typically lack the computational resources, high-speed networking interfaces, and modern electronic control units (ECUs) necessary to host LLM-based systems. Integrating such models into legacy platforms may require substantial retrofitting or reliance on edge/cloud offloading, both of which introduce additional cost, latency, and security considerations compared to modern vehicles equipped with advanced automotive SoCs.

**7) Misalignment and User Overtrust:** LLMs often sound confident even when wrong, which can lead to user overtrust—a well-documented risk in semi-autonomous systems. Drivers may defer to the LLM's response without verification, potentially reducing situational awareness. Aligning model behavior with human safety expectations, especially under uncertainty, remains a critical open research problem.

## VIII. EXISTING WORKS AND CASE STUDIES

**[26]** Liu et al. propose a trajectory prediction framework that exploits the extrapolation ability of off-the-shelf LLMs without fine-tuning. A frozen LLM, combined with a spatial encoder and reprogramming adapter, maps multi-agent traffic trajectories into the LLM's text embedding space for linear-decoder prediction. Evaluated on nuScenes, it captures complex maneuvers without HD maps, showing potential for planning and decision support. While competitive, it lags behind specialized models in fine-grained scene understanding, raising safety concerns. Security and privacy are not addressed, though robustness is emphasized as critical.

**[27]** Yang et al. present a comprehensive survey on LLMs for autonomous driving, reviewing applications, representative methods, and relevant datasets. The work addresses key security and privacy issues, noting risks such as privacy leakage, model illusions, adversarial inputs, and unsafe reasoning, and calls for rigorous ethical review before deployment.

**[28]** Du et al. propose a five-level proactivity framework for LLM-based in-vehicle conversational assistants, defined by intent assumption and action autonomy. The framework guides design and evaluation of proactive assistants but does not address security or privacy concerns.

TABLE II
THREAT LANDSCAPE OF IN-VEHICLE LLMS

| Threat Type | Description | Example |
|---|---|---|
| Prompt Injection | Malicious voice/text input manipulates LLM output | Triggering false diagnostics or overriding system decisions via phrasing |
| Model Inversion | Reconstruction of training or user data from responses | Revealing driver route history or diagnostics from LLM replies |
| Explainability Hallucination | LLM generates confident but incorrect causal explanations | Misleading cause of ABS brake triggering |
| Misalignment-Based Access | Overgeneralized interpretation of user intent bypasses access controls | "Let my friend drive" interpreted too broadly |

TABLE III
TAXONOMY OF IN-VEHICLE LLM APPLICATIONS

| LLM Role | Input Sources | Output Type | Deployment Level | Examples |
|---|---|---|---|---|
| Conversational Copilot | Voice, GPS, sensors | Spoken responses | Edge or Hybrid | "Should I enable lane assist now?" |
| Diagnostic Assistant | OBD-II codes, error logs | Human-readable explanation | Edge | "Why is the engine light on?" |
| Behavioral Coach | Acceleration, braking, steering logs | Driving feedback | Edge | "You're braking too late before exits" |
| Multimodal Summarizer | LiDAR, radar, camera, GPS | Summary text | Hybrid | "Vehicle on right is merging fast" |
| Technician Interface | Logs, voice commands | Maintenance report | Edge or Hybrid | "Relay fault codes to mechanic" |
| Infotainment Guide | Voice queries | Feature help / POI info | Cloud or Hybrid | "How to activate heated seats?" |

**[29]** Kong et al. propose a multi-agent LLM-based security framework for autonomous vehicles to prevent data leaks, ensure regulatory compliance, filter harmful queries, and validate outputs. Analyzed across eleven LLM-driven scenarios, the framework positions the LLM as a secure mediator between in-vehicle systems and cloud LLMs, ensuring privacy-preserving communication.

**[30]** Hamhoum et al. apply Mistral-7B in a cloud-edge misbehavior detection system for vehicular networks. Edge LLMs on roadside units perform real-time detection locally to preserve privacy, while cloud LLMs conduct deeper aggregated analysis. The design explicitly safeguards sensitive data and ensures regulatory compliance.

**[31]** Lu et al. present *OmniTester*, a multimodal LLM-based framework for generating realistic, diverse testing scenarios for autonomous vehicles. It improves adaptability and generalization over traditional methods, enhancing evaluation robustness, but does not address security or privacy concerns.

**[32]** Zhang et al. integrate LLAVA and deep reinforcement learning (GAE-PPO) to optimize data transmission and decision-making in vehicular networks. LLMs extract semantic features from visual sensor data, reducing raw data transmission by over 90%, while DRL adapts policies to bandwidth and QoE constraints. Simulations outperform PPO and DDPG baselines, but security and privacy are not addressed.

**[33]** Khiabani et al. optimize Small Language Models for in-vehicle deployment on resource-constrained hardware via pruning, healing, and quantization. These lightweight agents enable low-latency decision-making in vehicle control systems. Security and privacy are not discussed.

**[34]** Onsu et al. combine edge intelligence, 6G, and LLMs to enhance autonomous decision-making in Internet of Automated Defense Vehicles. LLMs support adaptive mission strategies and multimodal understanding in low-latency, resource-constrained environments. Cybersecurity is briefly noted, but the primary focus is on advancing autonomous defense capabilities.

**[35]** Orfanoudakis et al. integrate LLMs with graph neural networks to optimize EV smart charging. LLM-based Decision Transformers model sequences, while GNNs capture charging station relationships, improving satisfaction, grid compliance, and cost efficiency in simulations. Security and privacy are not addressed.

**[36]** Chiu et al. introduce the first Vehicle-to-Vehicle Question-Answering dataset and a baseline V2V-LLM model to enhance cooperative autonomous driving. The system improves multimodal reasoning for situational awareness but does not address security or privacy in cooperative communications.

**[37]** Onsu et al. combine LLaVA with YOLOv11-based instance segmentation for real-time multimodal traffic monitoring. LoRA fine-tuning and simulation-derived datasets improve query answering, achieving 84.3% location accuracy and 76.4% steering detection. Security and privacy are not considered.

**[38]** Lu et al. use LLMs for multi-lane convoy control, enabling adaptive maneuvers such as obstacle avoidance, vehicle joining/leaving, and escort switching. The method enhances convoy stability and flexibility but does not address security or privacy concerns.

**[39]** Cui et al. propose a multi-level vehicle–infrastructure cooperative decision-making framework for unsignalized intersections, combining infrastructure-level global planning with vehicle-level local adaptation for safer navigation. Security and privacy are not addressed.

**[40]** Yan et al. survey LLM applications in traffic and transportation, covering traffic prediction, vehicle control, ITS, and driver behavior modeling. LLM roles are classified as knowledge extractors, planners, and interactive agents, with discussion of datasets, benchmarks, and multimodal integration. Security and privacy are mentioned but not analyzed in depth.

**[41]** Zhang et al. present *LeAD*, a dual-rate driving architecture combining high-frequency E2E control with a low-frequency LLM planner for complex, long-tailed scenarios. The system improves scenario understanding and planning robustness but does not address security or privacy concerns.

**[42]** Yao et al. propose an LLM-agent framework for fine-grained, controllable traffic scenario generation, improving simulation realism and interpretability even with smaller LLMs. Security and privacy are not considered.

**[43]** Xu et al. optimize fairness and Age of Information in 5G NR V2I Mode 2 using an LLM-based multi-objective evolutionary algorithm for SPS window selection. The method balances fairness and timeliness but does not address security or privacy.

**[44]** Boateng et al. survey LLM applications in communication network and service management, categorizing device-, network-, and application-level use cases. Challenges include scalability, interpretability, and dataset scarcity. Security and privacy are noted but without in-depth technical analysis.

**[45]** Shoaib et al. survey frontier AI models, including LLMs, in Intelligent Transportation Systems for predictive modeling, decision automation, and multimodal fusion. The paper discusses integration challenges and explicitly notes security and privacy risks in ITS environments.

**[46]** Dhillon et al. explore LLM integration with on-board diagnostics for interpreting fault codes and generating repair recommendations, improving maintenance efficiency. Security and privacy are not explicitly addressed.

**[47]** Wandelt et al. review over 130 papers on LLM integration in ITS, covering domains such as autonomous driving, safety, tourism, traffic management, and human–machine interaction. Challenges include context-awareness, multimodal integration, fine-tuning, real-time responsiveness, and robustness. Security and privacy are noted as future work.

**[48]** Ma et al. propose AutoDH, combining LLMs with deep reinforcement learning to design heuristics for CVRP. Using a two-level MDP, the system dynamically selects or generates heuristics, outperforming traditional solvers and learning-based baselines. Security and privacy are not addressed.

**[49]** Alkhatib et al. present *CAN-BERT*, a transformer-based intrusion detection system for CAN bus traffic, detecting malicious messages with high accuracy. The work focuses on in-vehicle network security but does not address privacy.

**[50]** Tong et al. propose *ConnectGPT*, an LLM-powered pipeline for generating C-ITS safety messages from traffic observations, building on the ESRIUM project. Preliminary results show improved traffic management and safety, but broader security and privacy issues are not discussed.

**[51]** Sharma et al. design an LLM-based context-aware authentication framework for V2V/V2I communications, using situational data to prevent spoofing and unauthorized access. The work directly addresses security and privacy in V2X environments.

**[52]** Cui et al. survey multimodal LLM applications in autonomous driving, reviewing tools, datasets, and potential benefits for perception, planning, and interaction. Technical deployment challenges are discussed, but security and privacy are not addressed.

**[53]** Jafarnejad et al. present *KITT*, a voice assistant for transportation using Whisper, NexusRaven-V2, and XTTS-v2 for speech–LLM–speech interaction with real-time APIs. Security and privacy are not discussed.

**[54]** Wang et al. propose *Drive as Veteran*, a LoRA fine-tuned LLaMA-7B for generating driving decisions from natural language traffic descriptions. It generalizes well to varied scenarios but is limited in speed, complexity, and scope. Security and privacy are not addressed.

**[55]** Fu et al. explore LLMs for interpreting and reasoning about driving environments, assessing reasoning, memory, and adaptability. Security and privacy are not addressed.

**[56]** Xu et al. propose *DriveGPT4*, using GPT-4 for interpretable autonomous driving via a two-stage reasoning pipeline. It generates natural language descriptions and driving decisions from multimodal input. Security and privacy are not addressed.

**[57]** Rong et al. propose *STGLLM-E*, combining edge computing and GPT-2 for large-scale traffic flow prediction in 6G-IATS. Road networks are segmented via RoadSort, and subgraphs are processed with a spatio-temporal module and graph-enhanced LLM. Edge deployment with parameter transfer improves scalability and training efficiency, outperforming baselines. Security and privacy are not addressed.

**[58]** Xiong et al. assess LLM generalization for adaptive decision-making across five driving scenarios. Using the Dilu framework, LLMs outperform DQN and PPO in intersections and benefit from few-shot learning. Security and privacy are not considered.

**[59]** Wang et al. use LLMs for policy transfer by converting expert demonstrations into symbolic action sequences and grounding them into executable commands. The approach enables cross-domain adaptation with minimal fine-tuning, outperforming traditional methods. Security and privacy are not addressed.

**[60]** Gao et al. apply LLMs in multi-agent traffic simulation for scenario generation, cyberattack modeling, and defense

design in ITS. The framework highlights vulnerabilities such as traffic signal manipulation and false data injection, directly addressing cybersecurity challenges.

**[61]** This work explores GPT-3.5/4 for generating KQL queries from natural language to monitor network connectivity in connected vehicles. GPT-4 with context examples outperforms GPT-3.5, improving accessibility of network analytics. Security and privacy are not discussed.

**[62]** Mahmoudi et al. survey Generative AI for IoV intrusion detection, reviewing GAN-, VAE-, and Transformer-based methods for handling zero-day attacks and adversarial threats. Security is a central focus, with privacy noted in adversarial robustness context.

**[63]** Hernandez Salinas et al. present IDAS, a RAG-powered LLM voice assistant for querying vehicle manuals. Speech is transcribed, relevant content retrieved, and responses generated and spoken back. Security and privacy are not addressed.

**[64]** Furusawa et al. design an on-device LLM system for voice-based vehicle control and agent selection using Functionary on NVIDIA Jetson hardware. All processing remains local, significantly enhancing security and privacy.

**[65]** Fu et al. propose IoV-BERT-IDS, a BERT-based IDS using byte-sentence representation and novel pre-training tasks for robust intrusion detection across multiple vehicular datasets. Security is addressed; privacy is not a focus.

**[66]** Ge et al. envision an LLM-based operating system for automated vehicles, enabling modular, scalable task orchestration via natural language and multi-agent designs. Security and privacy are not addressed.

**[67]** Petrovic et al. use GPT-4 and LLaMA3 for automated test generation in CARLA, converting scenario descriptions into OCL rules and configurations. GPT-4 outperforms LLaMA3 in accuracy and speed, reducing testing time over tenfold. Security and privacy are not addressed.

**[68]** Honnalli et al. propose an LLM+RAG framework for anomaly detection in EV charging, identifying threats like billing fraud and tampering. Domain-specific retrieval improves detection accuracy over LSTM baselines. Security and privacy are directly addressed.

**[69]** K. et al. integrate LLMs into AD systems for improved reasoning, memory use, and handling of atypical driving scenarios. Experiments show enhanced adaptability over conventional frameworks. Security and privacy are not addressed.

**[70]** Xu et al. design an LLM-based Personalized Driver Agent delivering multimodal, driver-specific safety warnings using RAG for contextual reasoning. Security and privacy are not addressed.

**[71]** Murtaza et al. use ChatGPT to train drivers on ADAS/AV functions, outperforming traditional methods in speed, accuracy, and consistency. Security and privacy are not discussed.

**[72]** Zarz et al. combine LLMs, PCA, and deep learning for traffic accident prediction, improving situational awareness and proactive risk assessment. Security and privacy are not addressed.

**[73]** Du et al. propose a task-oriented semantic communication framework with LMMs that transmits only essential encoded features, reducing bandwidth and protecting visual privacy. Security and privacy are inherently addressed.

**[74]** Fang et al. present *CoDrivingLLM*, integrating simulation, reasoning, and memory for cooperative CAV driving in complex scenarios. Security and privacy are not addressed.

**[75]** Acharya et al. review LLM integration into AVs, highlighting benefits for end-to-end reasoning, planning, and scene comprehension, along with challenges in computation, alignment, and safety. Security and privacy are not addressed.

An abbreviated summary of all the papers has been highlighted in Table IV. While the integration of LLMs into intelligent vehicular systems offers compelling advances in reasoning, personalization, and automation, the field remains uneven in its attention to security and privacy as shown in Figure 3. Future research must prioritize trustworthy, secure-by-design approaches to make LLM-based vehicle intelligence suitable for real-world deployment.




Fig. 3. Distribution of Papers Based on Security and Privacy Concerns: A = Explicitly Addressed, B = Partially or Implicitly Addressed, C = Not Addressed

## IX. Research Gaps

From our analysis, several core limitations in current research emerge:

- **Lack of Embedded Deployment:** Most systems are evaluated on external hardware or rely heavily on cloud inference. Very few implement real-time, edge-resident LLMs suitable for ECU integration.
- **No Clear Interaction Architecture:** There is little consensus on how LLMs should interact with actuators, sensors, or CAN protocols in a safe and interpretable manner.
- **Security Neglect:** Only a handful of works discuss LLM-specific threats in automotive systems, such as prompt injection, misalignment, or OTA poisoning.
- **Explainability and Verification:** Most LLM outputs are uncontrolled and lack the verifiability required for safety-critical tasks.

- **No Standard Benchmarks:** Evaluation is mostly qualitative or dataset-limited, with no standard performance benchmarks tailored to in-vehicle LLM agents.
- **Limited Multimodal Reasoning:** Few systems explore sensor fusion (e.g., OBD-II + GPS + voice), which is crucial for context-aware in-car decisions.

**Actionable Research Questions:** The identified gaps raise several key questions for future work:

1) **Embedded Deployment:** How can LLMs run efficiently on in-vehicle embedded hardware under real-time, power, and thermal limits? What hardware–software co-designs enable ECU-level integration without cloud reliance?
2) **Interaction Architecture:** What is the safest interaction model between LLMs, sensors, actuators, and IVN protocols? How can boundaries prevent uncertified control?
3) **Security:** Which LLM-specific attacks (e.g., prompt injection, OTA poisoning) pose the most risk, and how can authentication and liveness detection be embedded by default?
4) **Explainability:** How can formal verification or runtime monitoring ensure safety compliance of LLM outputs? Which explainability methods fit high-stakes automotive use?
5) **Benchmarks:** What datasets and benchmarks best evaluate reasoning, latency, and robustness in multimodal in-vehicle contexts?
6) **Multimodal Reasoning:** How can heterogeneous inputs (voice, diagnostics, sensors) be fused efficiently for real-time decisions within resource limits?

## X. Future Research Directions

While the integration of Large Language Models (LLMs) into in-vehicle networks offers significant potential to enhance driver experience through advanced interaction, reasoning, and assistance capabilities, embedding such models within existing vehicular architectures remains a formidable challenge. In particular, seamless integration with legacy communication protocols such as the Controller Area Network (CAN) bus introduces constraints related to bandwidth, latency, and message prioritization, all of which can directly impact safety-critical operations. Addressing these challenges requires targeted research efforts that go beyond model development to encompass system-level considerations. The following subsections outline some of the key research directions aimed at overcoming these limitations and ensuring the safe, efficient, and trustworthy deployment of LLMs in real-world vehicular environments.

### *A. Voice-Driven Contextual LLM Agents in In-Vehicle Systems*

A promising future research direction lies in the design of voice-driven LLM agents that interface directly with both the driver and the in-vehicle network. In this framework, the LLM functions as a semantic mediator between natural language commands from the driver and structured signals from sensors and actuators. The user interacts with the system through voice—asking questions, issuing commands, or requesting status updates—while the LLM retrieves relevant real-time information from vehicle subsystems and formulates responses accordingly.

For example, a driver might ask, “Why is the check engine light on?”, “Can I reach my destination before it rains?”, or “What caused the lane change alert just now?” The LLM could access onboard diagnostics, GPS and map data, weather APIs, and ECU sensor values to generate context-rich answers. This would enable hands-free, personalized interaction and improve both situational awareness and driver trust. The system could also support proactive voice updates, such as “Your tire pressure is dropping rapidly” or “There is congestion 5 miles ahead on your current route.”

This framework requires the fusion of multiple data modalities—voice input, sensor telemetry, state estimates, and external sources—and poses challenges related to latency, hallucination control, and safety validation. However, with the emergence of edge-tuned LLMs and lightweight speech-to-text models, such a system could be deployed in a hybrid edge-cloud architecture. A modular implementation would also allow fallback to traditional rules when confidence in LLM output is low. The architecture of the voice-driven in-vehicle LLM framework is depicted in Figure 2.

We believe this represents a novel and impactful application space where LLMs are not just passive assistants but active interpreters of the driving context. Research is needed to define architectural boundaries, safety wrappers, fail-safe conditions, and privacy-preserving inference strategies for such agents in production vehicles.

### *B. Latency Evaluation for Network-Wide Integration*

Although LLMs are increasingly considered for real-time decision-making in vehicles, there is little empirical evidence quantifying their end-to-end latency once integrated into the complete in-vehicle network stack. Existing studies often benchmark LLM inference times in isolation, ignoring additional delays introduced by sensor fusion, CAN/FlexRay message propagation, safety monitoring, and actuator control loops. In real-time driving scenarios—such as collision avoidance or sudden rerouting—even sub-second delays can critically affect safety. Without systematic profiling under realistic conditions (e.g., different hardware accelerators, network loads, and environmental conditions), it is unclear whether LLM-driven functions can meet automotive real-time constraints or should be restricted to advisory, non-critical roles.

### *C. Lightweight LLM Architectures for Diverse Vehicle Classes*

State-of-the-art LLMs often require tens of gigabytes of memory and high-end GPUs, making them impractical for most production vehicles—especially in mixed-mode transportation systems that include older non-autonomous models and cost-sensitive semi-autonomous platforms. Current efforts in model quantization and distillation focus primarily on general-purpose AI benchmarks, not the constrained, safety-critical automotive domain. There is a research gap in designing domain-optimized, resource-aware LLMs that can run on embedded automotive-grade hardware without sacrificing accuracy in perception, reasoning, or driver interaction.

Achieving this would enable broader adoption of intelligent in-vehicle assistants across a heterogeneous fleet.

### D. Lightweight Authentication and Liveness Detection Mechanisms

If LLMs are to handle safety-relevant commands, the origin of prompts must be verified in real time to distinguish legitimate driver inputs from maliciously injected ones. While voice-based liveness detection [76] [77] [78] is a mature area in consumer devices, its direct application to in-vehicle LLMs is non-trivial: the vehicle environment introduces noise, variable microphone placement, and multi-speaker scenarios. Additionally, attacks such as *word stitching*—where an adversary assembles harmful commands from benign recorded fragments—remain underexplored in automotive contexts. Embedding continuous authentication and multimodal liveness checks (voice, touch, gaze) into the LLM pipeline, without introducing prohibitive latency, is an unresolved challenge requiring dedicated research.

### E. Formal Verification for Hallucination Mitigation

LLMs are prone to generating factually incorrect or contextually irrelevant outputs ("hallucinations"), which in a driving context could lead to unsafe or illegal actions. While formal verification techniques exist for control algorithms and perception models in autonomous vehicles, no equivalent frameworks exist for validating LLM outputs in safety-critical contexts. A formal method would mathematically guarantee that for a given valid input state, the LLM's output falls within an allowed action set, or otherwise triggers a fallback. Without such guarantees, even rare hallucinations could erode trust in the entire system. This research gap calls for novel verification approaches tailored to the probabilistic and generative nature of LLMs, potentially combining symbolic reasoning, constrained decoding, and runtime monitoring.

## XI. Conclusion

The integration of Large Language Models (LLMs) into in-vehicle systems marks a transformative shift in intelligent transportation, extending capabilities beyond traditional perception and control to include natural language interaction, context-aware reasoning, and explainable decision-making. This survey provides a structured taxonomy of LLM roles and deployment architectures, reviews key academic and industrial developments, and analyzes security and privacy considerations.

Our findings show rapid progress in applying LLMs to driver assistance, autonomous decision support, and multimodal understanding, yet critical challenges remain in achieving real-time performance, ensuring robustness against adversarial inputs, verifying outputs to avoid hallucinations, and maintaining trust and interpretability in safety-critical contexts. Addressing these gaps will require advances in efficient model design, secure integration, formal verification, and human–machine interaction.

This survey is timely as the automotive industry shifts toward software-defined vehicles, where LLM-enabled intelligence can significantly enhance safety, usability, and adaptability. By consolidating current knowledge, highlighting unresolved issues, and outlining research directions, this work provides a foundation for the safe, accountable, and effective deployment of LLMs in real-world driving environments.

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TABLE IV
CLUSTERED SUMMARY OF RELATED WORK BASED ON TYPE, CONTRIBUTION, AND SECURITY CONSIDERATION

| Type of Paper | Citation | Main Theme | Security/Privacy Addressed |
|---|---|---|---|
| Trajectory Prediction | [26] | LLMs for autonomous vehicle trajectory prediction | Not explicitly addressed, but robustness discussed |
| Survey | [27] | Survey of LLMs in autonomous driving, incl. challenges and datasets | Security and privacy extensively discussed |
| IV Conversational Assistant | [28] | LLMs for proactive in-vehicle conversational agents | Not addressed |
| Security Framework | [29] | Multi-agent LLM-based security framework for AVs | Security and privacy core focus |
| Misbehavior Detection | [30] | LLM-based cloud-edge system for misbehavior detection | Security and privacy addressed |
| Simulation & Testing | [31] | LLMs for generating diverse AV testing scenarios | Not addressed |
| Vehicular Networking | [32] | LLM+DRL for data efficiency and decision optimization | Not addressed |
| Resource Optimization | [33] | Deploying compressed SLMs in vehicles | Not addressed |
| Military Edge Intelligence | [34] | LLMs in edge intelligence for military vehicles | Security protocols touched upon |
| Smart Charging | [35] | LLMs+GNNs for smart EV charging | Not addressed |
| Cooperative Driving | [36] | LLM-driven vehicle-to-vehicle cooperative driving | Not addressed |
| Traffic Monitoring | [37] | LLaVA + YOLOv11 for intelligent traffic monitoring | Not addressed |
| Convoy Formation | [38] | LLM-based control for multi-lane convoy driving | Not addressed |
| Infrastructure Cooperation | [39] | Hierarchical vehicle-infrastructure decision-making | Not addressed |
| Survey | [40] | Survey of LLM use in transportation and traffic | Security/privacy not deeply analyzed |
| Imitation Learning | [41] | LLM + imitation learning dual-rate AD architecture | Not addressed |
| Scenario Augmentation | [42] | LLM-driven augmentation of driving scenarios | Not addressed |
| 5G/V2I Optimization | [43] | LLM + fairness/AoI optimization in V2I networks | Not addressed |
| Survey | [44] | Survey on LLMs in network/service management | Touches on security/privacy |
| Survey | [45] | Survey on LLMs in ITS and smart city infrastructure | Security/privacy discussed |
| Diagnostics | [46] | LLMs for vehicle diagnostics via fault interpretation | Not addressed |
| Survey | [47] | Review of 130+ papers on LLMs in ITS | No deep security/privacy analysis |
| Algorithm Design | [48] | LLM + DRL for CVRP heuristic generation | Not addressed |
| Intrusion Detection | [49] | CAN-BERT: LLM for CAN bus intrusion detection | Security focus, privacy not emphasized |
| C-ITS Messaging | [50] | LLMs for cooperative traffic safety messages | Not addressed |
| Authentication | [51] | LLM-based authentication in V2V/V2I | Security/privacy directly addressed |
| Survey | [52] | Survey on MLLMs for autonomous driving | Not addressed |
| Voice Assistant | [53] | LLM voice assistant for vehicle applications | Not addressed |
| Driving Decisions | [54] | LLM-generated human-like driving decisions | Not addressed |
| Reasoning Capacity | [55] | Reasoning and memory in driving scenarios | Not addressed |
| Interpretable Driving | [56] | DriveGPT4: reasoning explanations for driving | Not addressed |
| Edge + Traffic Prediction | [57] | STGLLM-E for scalable traffic prediction | Not addressed |
| Adaptive Policies | [58] | LLM adaptive policy learning across scenarios | Not addressed |
| Policy Transfer | [59] | LLM for symbolic policy transfer in AVs | Not addressed |
| Cybersecurity Testing | [60] | LLMs for traffic and cyberattack simulation | Security explicitly emphasized |
| Connectivity Monitoring | [61] | LLMs for natural language querying of connectivity | Not addressed |
| Survey | [62] | Survey on GenAI for IDS in IoV | Security focus, privacy acknowledged |
| Voice Query System | [63] | RAG + LLM for querying car manuals | Not addressed |
| On-Device AI | [64] | Edge LLMs for offline in-vehicle control | Security/privacy explicitly addressed |
| Hybrid IDS | [65] | IoV-BERT for hybrid IDS in IoV | Security addressed, privacy not central |
| OS Design | [66] | LLM as OS layer for vehicle control | Not addressed |
| Test Generation | [67] | LLMs for generating test cases (CARLA) | Privacy benefits noted |
| Anomaly Detection | [68] | RAG-enhanced LLM for EV charging anomaly detection | Security/privacy addressed |
| AV Integration | [69] | LLMs as cognitive layer in AV control | Not addressed |
| Personalized Driving Agent | [70] | LLMs for multimodal driver warnings | Not addressed |
| Training & Education | [71] | LLMs for ADAS/AV driver education | Not addressed |
| Traffic Prediction | [72] | LLM + DL for accident prediction | Not addressed |
| Semantic Communication | [73] | LMM-based semantic encoding in vehicle AI | Security/privacy directly addressed |
| Cooperative Driving | [74] | Interactive CoDrivingLLM for connected AVs | Not addressed |
| AV Architecture | [75] | LLMs as unified AV reasoning layer | Not addressed |